Pharmacy Distribution ERP — Database Schema & REST API Reference

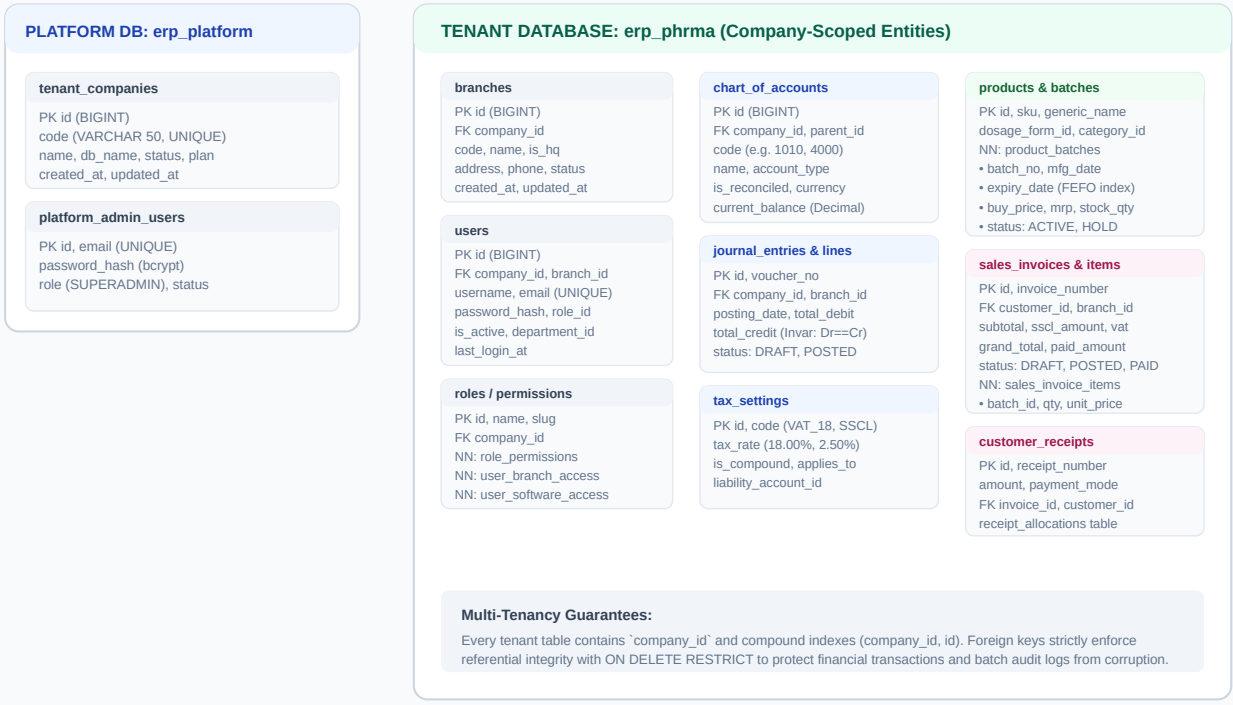
Document Code: `DOC-REF-2026-V1`
Target Audience: Backend Developers, DevOps Engineers, Database Administrators, System Integrators
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Backend Framework: Go 1.23 Gin REST API | **ORM:** GORM | **Database:** MySQL 8.0 Enterprise

1. Database Architecture & Schema Reference

The ERP utilizes a multi-tenant schema isolation architecture. The platform database (`erp_platform`) manages tenant lifecycle and subscription tiers, while operational tenant databases (`erp_phrma`) maintain company-scoped business entities with strict foreign key constraints.

Figure 4: Multi-Tenant Database Entity-Relationship Architecture

Platform Database Isolation, Tenant Core Entities, Financial Accounts, Inventory Batches & Sales Invoices



1.1 Platform Master Database Tables (`erp_platform`)

`tenant_companies`

Column	Type	Constraints	Description
`id`	`BIGINT UNSIGNED`	`AUTO_INCREMENT, PK`	Unique company identifier
`code`	`VARCHAR(50)`	`NOT NULL, UNIQUE`	Alphanumeric company code (e.g. `QA20261015`)
`name`	`VARCHAR(255)`	`NOT NULL`	Registered enterprise legal name
`legal_registration_no`	`VARCHAR(100)`	`NOT NULL`	Business registration number
`db_name`	`VARCHAR(100)`	`NOT NULL`	Target tenant database schema name
`status`	`VARCHAR(30)`	`NOT NULL, DEFAULT 'active'`	State: `active`, `suspended`, `pending`
`subscription_plan`	`VARCHAR(50)`	`NOT NULL`	Plan: `STANDARD`, `ENTERPRISE`
`created_at`	`DATETIME`	`NOT NULL`	Timestamp of provisioning
`updated_at`	`DATETIME`	`NOT NULL`	Timestamp of last modification

`platform_admin_users`

Column	Type	Constraints	Description
`id`	`BIGINT UNSIGNED`	`AUTO_INCREMENT, PK`	Platform administrator ID
`email`	`VARCHAR(255)`	`NOT NULL, UNIQUE`	Administrative login email
`password_hash`	`VARCHAR(255)`	`NOT NULL`	Salted bcrypt password hash
`role`	`VARCHAR(50)`	`NOT NULL, DEFAULT 'SUPERADMIN'`	Privilege level
`status`	`VARCHAR(30)`	`NOT NULL, DEFAULT 'active'`	Status: `active`, `inactive`

1.2 Tenant Database Tables (`erp_phrma`)

`companies` & **`branches`**

- `companies`: Stores company settings, base currency (`LKR`), timezone, logo URL.
- `branches`:
- `id` (`BIGINT UNSIGNED PK`)
- `company_id` (`BIGINT UNSIGNED NOT NULL, FK -> companies.id`)

- `code` (`VARCHAR(50) NOT NULL`), `name` (`VARCHAR(255) NOT NULL`)
- `is_hq` (`BOOLEAN NOT NULL DEFAULT FALSE`)
- `status` (`VARCHAR(30) NOT NULL DEFAULT 'active'`)

`users`, `roles`, & `permissions`

- `users`: `id`, `company_id`, `branch_id`, `username`, `email`, `password_hash`, `is_active`, `department_id`, `designation_id`.
- `roles`: `id`, `company_id`, `name`, `code`, `description`.
- `permissions`: `id`, `name`, `slug` (e.g. `finance:journal:post`, `inventory:grn:approve`).
- `role_permissions`: Composite PK `(role_id, permission_id)`.
- `user_branch_access`: Composite PK `(user_id, branch_id)`.
- `user_software_access`: Composite PK `(user_id, module_id)`.

`chart_of_accounts`

Column	Type	Constraints	Description
`id`	`BIGINT UNSIGNED`	`AUTO_INCREMENT, PK`	Account identifier
`company_id`	`BIGINT UNSIGNED`	`NOT NULL, FK`	Scoped tenant ID
`code`	`VARCHAR(50)`	`NOT NULL`	Account code (e.g. `1010`, `1100`, `4000`)
`name`	`VARCHAR(255)`	`NOT NULL`	Account title
`parent_id`	`BIGINT UNSIGNED`	`NULL, FK -> chart_of_accounts.id`	Tree hierarchy parent account
`account_type`	`VARCHAR(50)`	`NOT NULL`	`ASSET`, `LIABILITY`, `EQUITY`, `REVENUE`, `EXPENSE`
`is_reconciled`	`BOOLEAN`	`DEFAULT FALSE`	Reconciliation flag
`current_balance`	`DECIMAL(18,4)`	`NOT NULL, DEFAULT 0.0000`	Real-time computed balance

`journal_entries` & `journal_entry_lines`

- `journal_entries`: `id`, `company_id`, `branch_id`, `financial_year_id`, `voucher_no`, `posting_date`, `total_debit`, `total_credit`, `status` (`DRAFT`, `POSTED`, `CANCELLED`).
- `journal_entry_lines`: `id`, `journal_entry_id`, `account_id`, `debit_amount`, `credit_amount`, `description`.
- **Database Constraint:** Check constraint / Application trigger verifying $\text{total_debit} == \text{total_credit}$.

`products` & `product_batches`

- `products`: `id`, `company_id`, `sku`, `generic_name`, `brand_name`, `dosage_form_id`, `category_id`, `base_unit_id`, `min_shelf_life_days`, `is_taxable`.
- `product_batches`: | Column | Type | Constraints | Description | |---|---|---|---| | `id` | `BIGINT UNSIGNED` | `AUTO_INCREMENT, PK` | Unique batch ID | | `company_id` | `BIGINT UNSIGNED` | `NOT NULL, FK` | Scoped tenant ID | | `product_id` | `BIGINT UNSIGNED` | `NOT NULL, FK` | Foreign key to products | | `batch_no` | `VARCHAR(100)` | `NOT NULL` | Physical lot/batch string | | `mfg_date` | `DATE` | `NOT NULL` | Date of manufacture | | `expiry_date` | `DATE` | `NOT NULL, INDEX` | Expiry date (FEFO sort key) | | `buy_price` | `DECIMAL(18,4)` | `NOT NULL` | Purchase cost per unit | | `mrp` | `DECIMAL(18,4)` | `NOT NULL` | Maximum retail price | | `stock_quantity` | `DECIMAL(18,4)` | `NOT NULL, DEFAULT 0` | Current physical stock | | `status` | `VARCHAR(30)` | `NOT NULL, DEFAULT 'ACTIVE'` | Status: `ACTIVE`, `HOLD`, `EXPIRED` |

`sales_invoices` & `sales_invoice_items`

- `sales_invoices`: `id`, `company_id`, `branch_id`, `customer_id`, `invoice_number`, `invoice_date`, `subtotal`, `sscl_amount`, `vat_amount`, `grand_total`, `paid_amount`, `status` (`DRAFT`, `POSTED`, `PAID`, `CANCELLED`).
- `sales_invoice_items`: `id`, `sales_invoice_id`, `product_id`, `batch_id`, `quantity`, `unit_price`, `subtotal`, `tax_amount`, `total_amount`.

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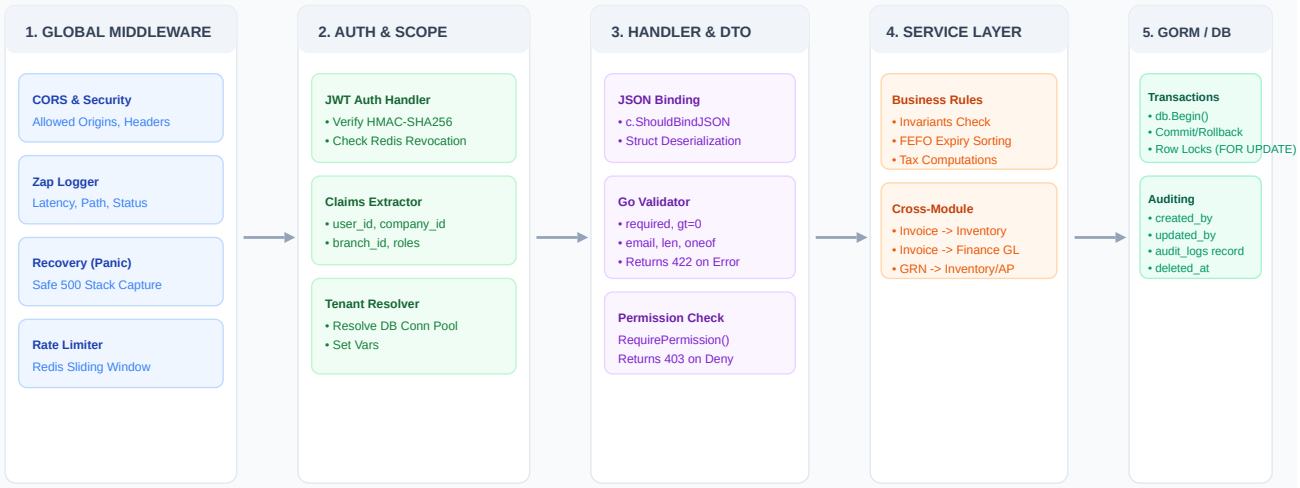
2. Complete REST API Endpoint Reference

All endpoints expect JSON request payloads and return standard envelope responses:

```
{
  "success": true,
  "data": { ... },
  "message": "Operation completed successfully"
}
```

Figure 3: Go Gin Backend Request-Processing Pipeline

Comprehensive Middleware Lifecycle, DTO Validation, Service Business Logic & GORM Database Transactions



2.1 Authentication & Platform Administration APIs

Method	Endpoint	Permission Required	Description
`POST`	`/api/v1/auth/login`	*Public*	Authenticates company user, returns JWT and user profile
`POST`	`/api/v1/auth/logout`	*Authenticated*	Invalidates JWT and places token JTI in Redis blacklist
`POST`	`/api/v1/platform-admin/login`	*Public*	Authenticates global platform administrator
`GET`	`/api/v1/platform-admin/companies`	*Platform Admin*	Lists all registered tenant companies
`POST`	`/api/v1/platform-admin/companies`	*Platform Admin*	Provisions a new tenant company database and admin user

2.2 Control Center APIs

Method	Endpoint	Permission Required	Description
`GET`	`/api/v1/control/company`	`control:company:read`	Returns active company profile and settings
`PUT`	`/api/v1/control/company`	`control:company:write`	Updates active company profile
`GET`	`/api/v1/control/branches`	`control:branch:read`	Lists branches belonging strictly to current tenant
`POST`	`/api/v1/control/branches`	`control:branch:write`	Creates new organizational branch
`GET`	`/api/v1/control/users`	`control:user:read`	Lists tenant users with branch and role assignments
`POST`	`/api/v1/control/users`	`control:user:manage`	Creates a new tenant user via the wizard
`GET`	`/api/v1/control/roles`	`control:role:read`	Lists enterprise roles
`GET`	`/api/v1/control/roles/:id/permissions`	`control:role:read`	Gets permission matrix for specified role
`PUT`	`/api/v1/control/roles/:id/permissions`	`control:role:write`	Updates role permission assignments
`GET`	`/api/v1/control/access-audit`	`control:audit:view`	Retrieves system security access audit logs

2.3 Finance APIs

Method	Endpoint	Permission Required	Description
`GET`	`/api/v1/finance/chart-of-accounts`	`finance:coa:view`	Returns hierarchical chart of accounts tree
`POST`	`/api/v1/finance/chart-of-accounts`	`finance:coa:edit`	Creates a new general ledger account
`GET`	`/api/v1/finance/journal-entries`	`finance:journal:view`	Lists journal entry vouchers
`POST`	`/api/v1/finance/journal-entries`	`finance:journal:create`	Creates a draft journal entry
`POST`	`/api/v1/finance/journal-entries/:id/post`	`finance:journal:post`	Verifies balance invariant and posts to general ledger
`GET`	`/api/v1/finance/tax-settings`	`finance:tax:view`	Lists configured VAT and SSCL tax rules
`POST`	`/api/v1/finance/tax-settings`	`finance:tax:manage`	Configures tax rates, compounds, and liability accounts

2.4 Inventory & Batch APIs

Method	Endpoint	Permission Required	Description
`GET`	`/api/v1/inventory/products`	`inventory:product:read`	Lists catalog products with dosage forms and categories
`POST`	`/api/v1/inventory/products`	`inventory:product:create`	Adds new pharmaceutical SKU
`GET`	`/api/v1/inventory/product-batches`	`inventory:batch:read`	Lists batches with expiry dates and stock balances
`GET`	`/api/v1/inventory/product-batches/fefo`	`inventory:batch:read`	Returns batches sorted strictly in FEFO priority order
`POST`	`/api/v1/inventory/product-batches/:id/quarantine`	`compliance:batch:quarantine`	Quarantines batch, switching status to `HOLD`
`GET`	`/api/v1/inventory/grns`	`inventory:grn:view`	Lists Goods Receipt Notes
`POST`	`/api/v1/inventory/grns`	`inventory:grn:create`	Creates draft GRN with batch and expiry details
`POST`	`/api/v1/inventory/grns/:id/approve`	`inventory:grn:approve`	Approves GRN, increments stock, writes stock ledger
`GET`	`/api/v1/inventory/stock-ledger`	`inventory:ledger:view`	Queries append-only stock movement audit ledger

2.5 Invoice Center APIs

Method	Endpoint	Permission Required	Description
`GET`	`/api/v1/invoice-center/customers`	`invoice:customer:read`	Lists customers with credit limits and balances
`POST`	`/api/v1/invoice-center/customers`	`invoice:customer:create`	Creates customer record
`GET`	`/api/v1/invoice-center/sales-orders`	`invoice:order:read`	Lists sales orders
`POST`	`/api/v1/invoice-center/sales-orders`	`invoice:order:create`	Creates draft sales order
`POST`	`/api/v1/invoice-center/sales-orders/:id/approve`	`invoice:order:approve`	Validates credit limit and reserves FEFO batch stock
`GET`	`/api/v1/invoice-center/sales-invoices`	`invoice:bill:read`	Lists sales invoices
`POST`	`/api/v1/invoice-center/sales-invoices`	`invoice:bill:create`	Creates draft sales invoice
`POST`	`/api/v1/invoice-center/sales-invoices/:id/post`	`invoice:bill:issue`	Deducts inventory, posts balanced GL journals, finalizes invoice
`GET`	`/api/v1/invoice-center/customer-receipts`	`invoice:receipt:read`	Lists customer receipt payments
`POST`	`/api/v1/invoice-center/customer-receipts`	`invoice:receipt:collect`	Records customer payment and settles invoice balance

2.6 Compliance Center APIs

Method	Endpoint	Permission Required	Description
`GET`	`/api/v1/compliance-center/recalls`	`compliance:recall:view`	Lists statutory product batch recalls
`POST`	`/api/v1/compliance-center/recalls`	`compliance:recall:create`	Initiates formal Class I / Class II recall
`GET`	`/api/v1/compliance-center/recalls/:batch_id/trace`	`compliance:recall:view`	Generates customer recall audit trace list
`POST`	`/api/v1/compliance-center/disposals`	`compliance:disposal:manage`	Records drug destruction certificate and GL write-off